

Saminur Karim

When I opened the encrypted document, it was filled with random keys and letters. Multiple lines of “^” were shown and most of the document was encrypted. There were multiple hashing and encryption algorithms used. Looking at the metadata, the cryptographic protocols used were listed. Here is a copy and paste of the listed algorithms. cipherAlgorithm="AES" cipherChaining="ChainingModeCBC" hashAlgorithm="SHA512" saltValue="tkYzyyS. SHA512 used a 512 byte for the hash algorithm which detects the integrity and non-repudiation of the document. AES encryption used a 256-bit size key. A salt was also added to protect the document's hash, so it is not repeatable. Overall, these are strong encryption protocols, but they may still have weaknesses. Key mismanagement, weak passwords, and metadata exposure as well. SHA-512 is a good encryption algorithm however it is susceptible to brute force attacks. It is necessary to salt the hash but having a weak one will make it easier to crack. SHA-512 is not good for short string documents as with accelerated GPUs it can be cracked. Overall, if a weak password is used than the word document will be easily cracked. Word documents derive keys that turn passwords into keys as well as salting the password.

```

! "# $ % & .....
Root
Entry .....
`S'Ø<ÄEncryptedPackage"..... H7DataSpaces..... ØR'Ø<ÏR
'Ø<Version..... L.....

<Microsoft.Container.DataSpacesh
EncryptedPackage2StrongEncryptionDataSpace2StrongEncryptionTransformXL{FF9A3F03-56EF-4613-
BDD5-5A41C1D07246}NMicrosoft.Container.EncryptionTransformDataSpaceMap..... pDataSpaceInfo..... ØR
'Ø<ØR'Ø<StrongEncryptionDataSpace4..... @TransformInfo..... ØR'Ø<ΔR
'Ø<StrongEncryptionTransform4..... ØR'Ø<ΔR'Ø<Primary..... »EncryptionInfo.....
@<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<encryption xmlns="http://schemas.microsoft.com/office/2006/encryption" xmlns:p="http://
schemas.microsoft.com/office/2006/keyEncryptor/password" xmlns:c="http://schemas.microsoft.com/
office/2006/keyEncryptor/certificate"><keyData saltSize="16" blockSize="16" keyBits="256"
hashSize="64" cipherAlgorithm="AES" cipherChaining="ChainingModeCBC" hashAlgorithm="SHA512"
saltValue="tkYzvvs

```

